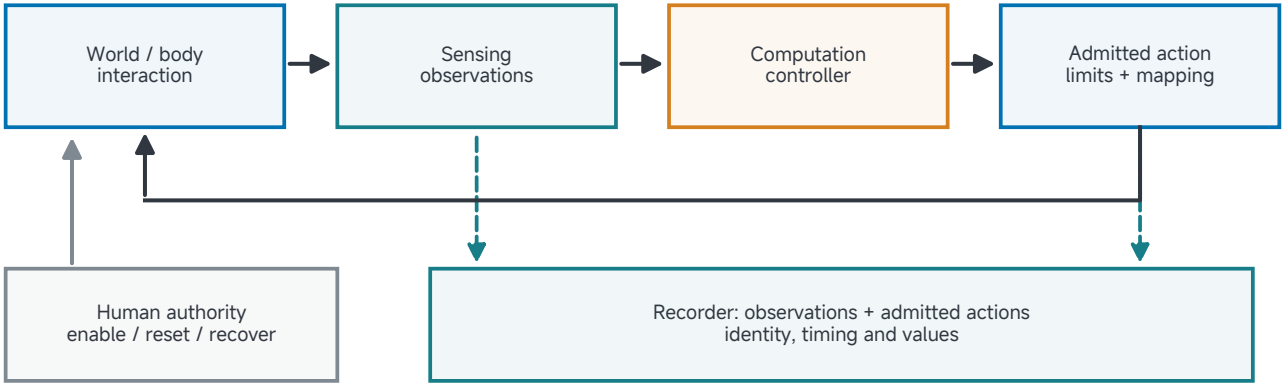Solid loop: action and world response